

Fig. 2: An actual-size, single-side PCB for the door lock circuit

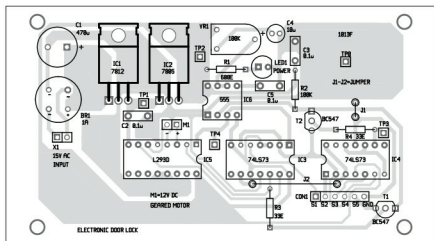


Fig. 3: Component layout for the PCB

flop FF2 of IC4 toggles from logic 0 to 1. This provides a positive bias to the base of transistor T2.

Now when switch S4 is pressed, the emitter of T2 is pulled to ground momentarily. At this point, transistor T2 is driven into saturation. This clears all the flip-flops and triggers 555 timer (IC6) simultaneously.

Timer IC6 is configured in monostable mode. It provides an output pulse of predefined width, which simultaneously clocks flip-flop FF1 of

IC3 and enables the H-bridge. This rotates the motor for the predefined period to unlock the door.

To lock the door back again, just press switch S5. This triggers IC6, which, in turn, toggles flip-flop FF1 of IC3, rotating the motor in reverse direction for the same period of time.

Working of the project is simple. When the keys are pressed in the sequence S1-S2-S3-S4, the door unlocks. These keys can be connected to any numbers on the keypad, thus making

PARTS LIST

Semiconductors:

- | | |
|--|---|
| IC1 | - 7812, 12V regulator |
| IC2 | - 7805, 5V regulator |
| IC3, IC4 | - 74LS573, dual J-K flip-flop |
| IC5 | - L293D H-bridge driver |
| IC6 | - 55 timer |
| BR1 | - 1A bridge rectifier module |
| T1, T2 | - BC147 npn transistor |
| LED1 | - 5mm LED |
| Resistors (all $\frac{1}{4}$ -watt, $\pm 5\%$ carbon): | |
| R1 | - 680-ohm |
| R2 | - 100-kilo-ohm |
| R3, R4 | - 33-ohm |
| VR1 | - 100-kilo-ohm preset |
| Capacitors: | |
| C1 | - 470 μ F, 35V electrolytic |
| C2, C3, C5 | - 0.1 μ F ceramic |
| C4 | - 10 μ F, 16V electrolytic |
| Miscellaneous: | |
| CON1 | - 6-pin connector |
| SI-S5 | - Tactile switches for 4x5 keypad |
| M1 | - 12V, 10rpm geared DC motor |
| X1 | - 230V AC primary to 15V, 500mA secondary transformer |

your password. To lock the door, just press S5.

Construction and testing

An actual-size, single-side PCB for the electronic door lock circuit is shown in Fig. 2 and its component layout in Fig. 3. Assemble the circuit on the PCB to minimise assembly errors. Take any 4x5 numeric keypad and connect five of its keys to the PCB at CON1 (refer Fig. 1 for connections). Please note, pressing the right sequence of keys corresponding to S1-S2-S3-S4 is necessary for unlocking the door.

To test the circuit for proper functioning, verify 12V power supply at TP1 and 5V power supply at TP2 with respect to TP0. Press S1 through S3 sequentially and check high level at TP3. Now press S4 and check the output of IC6. It should be high for a time period decided by the timing components. ●

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